Computer Networks
CL3001

LAB – 04
Routers
&
Telnet, SSH

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Computer Networks Lab 04

Course: Computer Networks (CL3001) **Semester:** Spring 2026
Instructor: Sameer Faisal **T.A:** N/A

Note:
- Maintain discipline during the lab.
- Listen and follow the instructions as they are given.
- Just raise hand if you have any problem.
- Completing all tasks of each lab is compulsory.
- Get your lab checked at the end of the session.

Lab Objective

- Introduction to Router and basic Configuration
- Introduction to Telnet & configuration of Telnet in Cisco Packet Tracer.
- Introduction to SSH & configuration of SSH in Cisco Packet Tracer.

INTRODUCTION TO ROUTERS

A router is a networking device that connects multiple computer networks and manages the transfer of data between them through wired or wireless connections.

1. How Routers Work?

In technical terms, a router is a Layer 3 network gateway device, meaning that it connects two or more networks and that the router operates at the network layer of the OSI model.
Routers contain a processor (CPU), several kinds of digital memory, and input-output (I/O) interfaces. They function as special-purpose computers, one that does not require a keyboard or display. The router's memory stores an embedded operating system (OS). Compared to general-purpose OS products like Microsoft Windows or Apple Mac OS, router operating systems limit what kind of applications can be run on them and also need much smaller amounts of storage space. Examples of popular router operating systems include Cisco Internetwork Operating System (IOS) and DD-WRT. These operating systems are manufactured into a binary firmware image and are commonly called router firmware.
By maintaining configuration information in a part of memory called the routing table, routers also can filter both incoming or outgoing traffic based on the addresses of senders and receivers.
Routers became mainstream consumer devices when households began to accumulate multiple computers and wanted to share the home Internet connection Home networks use Internet Protocol (IP) routers to connect computers to each other and to the Internet. Early generations of home routers supported wired

networking with Ethernet cables while newer wireless routers supported Wi-Fi together with Ethernet. The term broadband router applies to any home wired or wireless router being used for sharing a broadband Internet connection.

2. Routing Table

A routing table contains the information necessary to forward a packet along the best path toward its destination. Each packet contains information about its origin and destination. When a packet is received, a network device examines the packet and matches it to the routing table entry providing the best match for its destination. The table then provides the device with instructions for sending the packet to the next hop on its route across the network.

A basic routing table includes the following information:

- **Destination**: The IP address of the packet's final destination.
- **Next hop:** The IP address to which the packet is forwarded.
- **Interface:** The outgoing network interface the device should use when forwarding the packet to the next hop or final destination.

There is much more information present in the routing table which will be discussed in the upcoming labs.

3. Simple Router Configuration

In this lab we will only have a look at how we can assign IPs to router interfaces using GUI (Graphical User Interface) and CLI (Command Line Interface). Further configurations will be done in later labs.

IP Assignment:

Following is the way for IP assignment to a router interface (port) using CLI.

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface e 0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shut
%LINK-3-UPDOWN: Interface Ethernet0, changed state to up
Router(config-if)#exit
Router(config)#exit
```

SSH & Telnet

1. Introduction to Telnet

A terminal emulation program for TCP/IP networks such as the Internet. The Telnet program runs on your computer and connects your PC to a server on the network. You can then enter commands through the Telnet program and they will be executed as if you were entering them directly on the server console. This enables you to control the server and communicate with other servers on the network. To start a Telnet session, you must log in to a server by entering a valid username and password. Telnet is a common way to remotely control Web servers.

To telnet means to establish a connection with the Telnet protocol, either with command line client or with a programmatic interface.

2. Configuration of Telnet

Below are the steps for Telnet Protocol. Follow the procedure for the configuration of Telnet Protocol.

Step 1: Build the following topology.

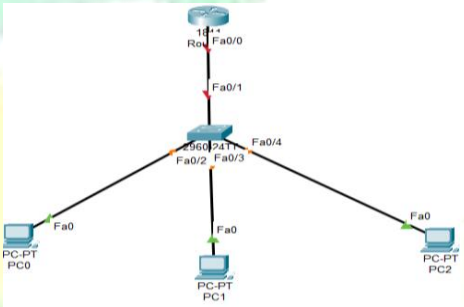

Figure 1

Step 2: Set IPs on the PCs. As, by default, all PCs are in vlan. We will create a virtual interface on switch with vlan 1 as follows:

```
Switch>en
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.1.1 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
```

Figure 2

Assigning IP to PCs:

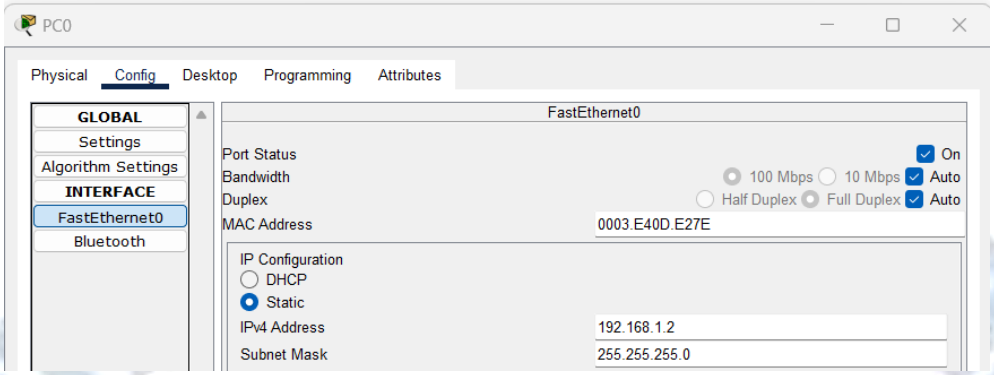

Figure 3

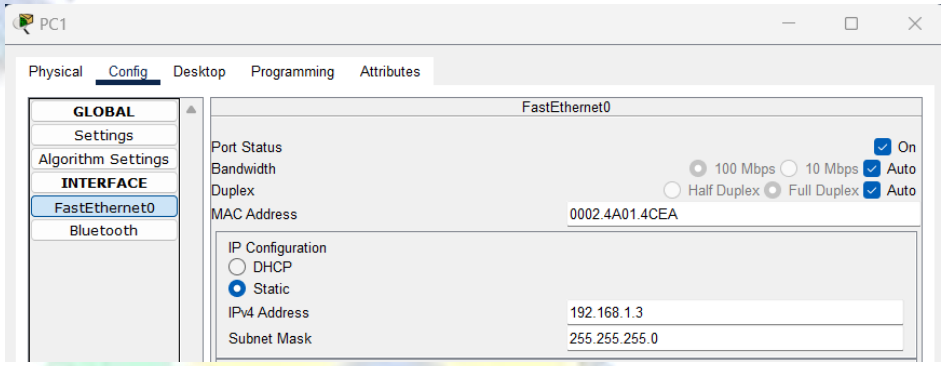

Figure 4

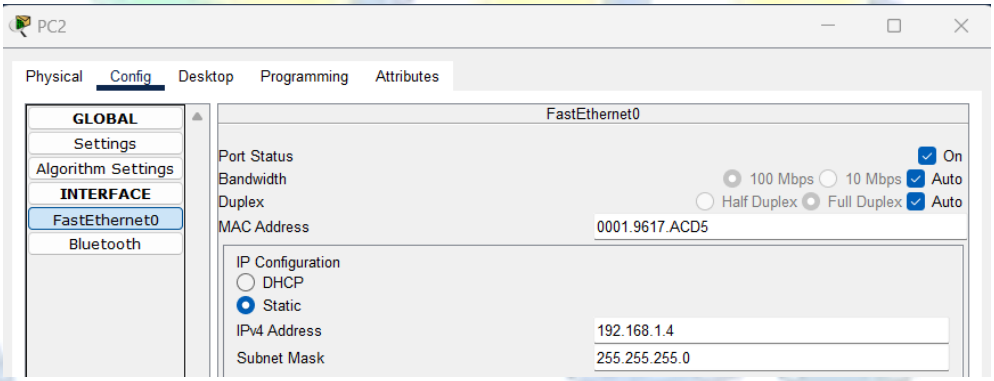

Figure 5

Step 3: Now, try to telnet the switch from our PC, it refuses because we have not applied authentication on the switch yet.

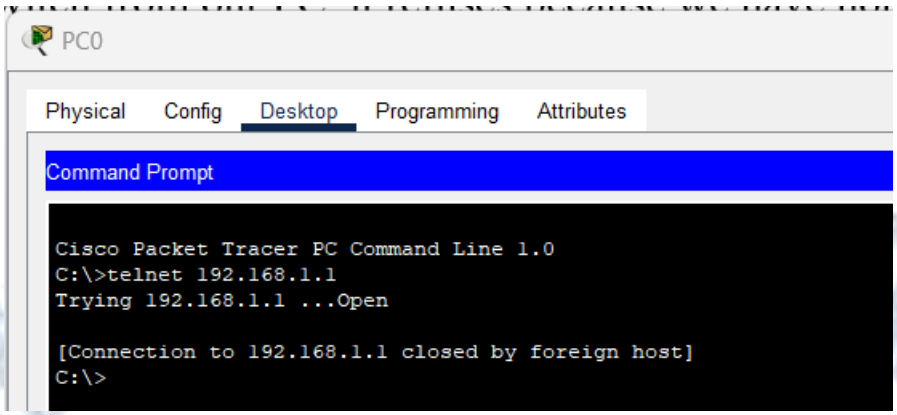

Figure 6

Secure Shell Server (SSH) and FTP services. Each Telnet, SSH, or FTP session requires one vty-line. You can add security to your system by configuring the software to validate login requests:

```
Switch(config-if)#exit
Switch(config)#line vty 0 15
Switch(config-line)#password cisco
Switch(config-line)#login
Switch(config-line)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

Figure 7

Now, we can easily telnet. But it does not let us go in the switch enabled mode because we have not set the password on the switch yet.

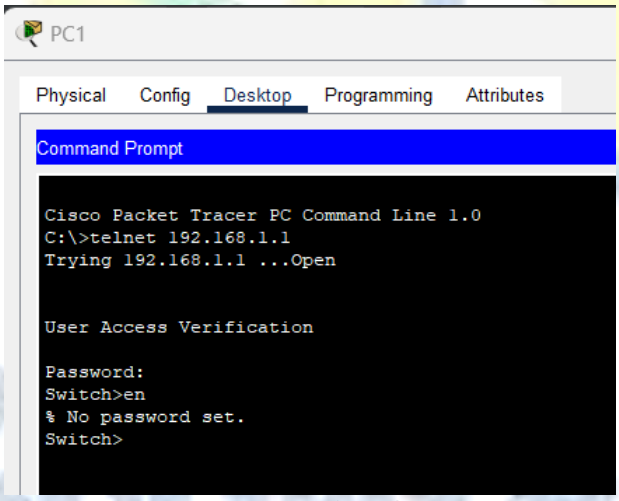

Figure 8

Let’s apply password on the switch enabled mode.

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#enable password cs
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Figure 9

Now, we can go inside Switch configuration mode from our pc.

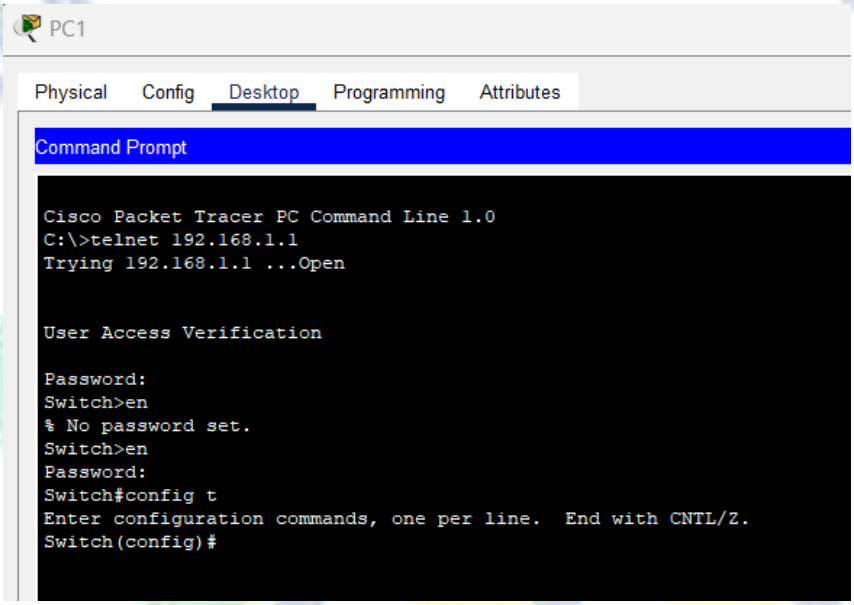

Figure 10

3. Introduction to SSH

Secure Shell or Secure Socket Shell is a network protocol. It is an application layer protocol that is in the 7th later of the Open Systems Interconnection (OSI) network model. It also refers to the suite of utilities that implements the SSH protocol.

Secure Shell also supports both password and key-based authentication. Password-based authentication let users provide username and password to authenticate to the remote server.

A key-based authentication allows users to authenticate through a key-pair. The key pairs are two cryptographically secure keys for authenticating a client to a Secure Shell server.

Furthermore, the Secure Shell protocol also encrypts data communication between two computers. It is extensively used to communicate with a remote computer over the Internet.

4. Configuration of SSH

Taking the same topology as mentioned in telnet.

Below are the steps for SSH Protocol. Follow the following procedure for the configuration of SSH Protocol.

```
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname sameer
sameer(config)#ip domain name ai-study
sameer(config)#crypto key generate rsa
The name for the keys will be: sameer.ai-study
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

sameer(config)#
```

Figure 11

```
sameer(config)#ip s
*Mar 1 0:29:49.71: %SSH-5-ENABLED: SSH 1.99 has been enabled
% Ambiguous command: "ip s"
sameer(config)#ip ssh version 2
sameer(config)#line vty 0 15
sameer(config-line)#transport input ssh
sameer(config-line)#
```

Figure 12

Protocol working on it. By default, username is admin. We can apply any sort of configuration on our switch from our PC.

```
C:\>ssh -l admin 192.168.1.1

Password:

Password:



sameer>en
Password:
sameer#config t
Enter configuration commands, one per line.  End with CNTL/Z.
sameer(config)#interface fa0/2
sameer(config-if)#no shutdown
sameer(config-if)#exit
sameer(config)#exit
sameer#
```

Figure 13

Now, if we want to change the username from admin to something else, we will do it as follows:

```
sameer(config)#username ds-study secret abc
sameer(config)#line vty 0 15
sameer(config-line)#login local
sameer(config-line)#
```

Figure 14

Now we will do the following on our PCs:

```
C:\>ssh -l ds-study 192.168.1.1

Password:
% Login invalid


Password:
% Login invalid


Password:



sameer>
```

Figure 15